

Modularity of elliptic curves over CM and totally real fields

AI-assisted research draft*

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Abstract

We assemble the field extension of the geometric prime switch in `openai/math`, family 030, and a proposed treatment of its remaining characteristic-five obstruction. The source’s explicit geometric and local inputs imply modularity over every CM field not containing a primitive fifth root of unity, with no Galois or degree restriction. Quadratic descent then gives modularity over every totally real field. For CM fields containing fifth roots of unity, a split Tate condition with parameter valuation prime to five excludes the exceptional adjoint cohomology class. We give a partially framed ordinary patching argument using the local rings of Allen–Khare–Thorne. The width-three second degeneration of the geometric family produces precisely this Tate condition. Validation of the new patching argument would complete the extension to all CM fields. Independently of the geometric construction, it would remove the fifth-root-of-unity restriction from Caraiani–Newton’s residual-mod-three modularity criterion.

Status. This is a research draft. The package $\mathcal{G}$ in Section 2 records the geometric and local assertions used from family 030; that source manuscript is not independently certified here. The argument $\mathcal{T}_5$ in Proposition 10.1 is a new patching adaptation presented for independent review. The CM theorem without fifth roots of unity and the totally real theorem use $\mathcal{G}$ only. The all-CM extension uses both $\mathcal{G}$ and $\mathcal{T}_5$.

1 Main statements and conventions

A CM field is a totally imaginary quadratic extension of a totally real number field. It need not be Galois over $\mathbb{Q}$. For an elliptic curve E/K we write

$$V_\ell(E) = H_{\text{ét}}^1(E_{\overline{K}}, \mathbb{Q}_\ell), \quad \det V_\ell(E) = \epsilon_\ell^{-1},$$

with labelled Hodge–Tate weights $\{0, 1\}$. We use weight zero for the cohomological automorphic normalization; over a totally real field this means parallel classical weight two. The notation $E[\ell]$ uses the usual Tate-module convention. Passing to its dual does not change any residual irreducibility, image-up-to-conjugacy, or decomposed-generic condition used below.

For a non-CM elliptic curve, modularity means a cuspidal weight-zero automorphic representation with its Galois representations and finite local parameters. For a CM elliptic curve we use the classical Hecke-character construction: its realization is an isobaric sum when the CM field is contained in the ground field, and quadratic automorphic induction otherwise. Thus cuspidality is not asserted for a split CM realization.

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For a residual representation of characteristic ℓ , decomposed genericity means that there is a rational prime $q \neq \ell$, completely split in the ground field, such that the representation is unramified at every place above q and every ratio of two distinct indexed Frobenius eigenvalues differs from q modulo ℓ ; see [CN], Definition 2.1.27. We will always construct witnesses with $q \equiv 1 \pmod{\ell}$ and two distinct eigenvalues.

Here are the conclusions and their dependencies. The proofs appear in the indicated sections; the input symbols are defined there without assuming any all-CM or all-totally-real modularity statement.

Conclusion	Inputs	Location
All elliptic curves over CM K with $\zeta_5 \notin K$	$\mathcal{G}$	Theorem 5.1
All elliptic curves over totally real fields	$\mathcal{G}$	Corollary 6.1
All elliptic curves over every CM field	$\mathcal{G} + \mathcal{T}_5$	Corollary 13.1
Residual-mod-three modularity over every CM field	$\mathcal{T}_5$	Corollary 14.1

The last row retains decomposed genericity and absolute irreducibility over the field obtained by adjoining ζ_3 ; it is not an unrestricted modularity claim. In particular the totally real conclusion is logically independent of the proposed exceptional-five step.

The two field choices driving the argument are different. First prove the auxiliary elliptic curve modular over its parameter's ground field; only then adjoin the cyclotomic characters for the three congruences. For a totally real ground field, pass to a CM quadratic extension chosen to avoid ζ_5 and descend. For the remaining CM case, keep the characteristic-five seed and control its local ordinary deformation condition geometrically.

2 Geometric systems and local inputs

We write $G_L = \text{Gal}(\overline{L}/L)$ and use the cohomological convention $\det H_{\text{et}}^1(E, \mathbb{Q}_\ell) = \epsilon_\ell^{-1}$, with labeled Hodge–Tate weights $\{0, 1\}$. The following package, denoted $\mathcal{G}$, records the source inputs. The source [M] is the manuscript dated 4 October 2026 listed in the bibliography; its assertions are used in precisely the forms stated below.

G1. Covers and compatible systems. For an elliptic curve E/k over a number field and odd n , [M], Lemma 3.1 constructs $U \subset \mathbb{A}_k^1$, quadratic twists B_t of E , and smooth genus- m curves $C_m(t)$ for $m \mid n$. The covers have rotations after adjoining μ_m , and an involution w over k inverting those rotations. For $n = 3p$, the character pieces V_θ over $F = k(\mu_{3p})$ are rank-two compatible systems. Proposition 5.1 gives semisimplicity, determinant ϵ_ℓ^{-1} , labelled Hodge–Tate weights $\{0, 1\}$ at every embedding, potential semistability, and compatible Frobenius polynomials. The construction is over arbitrary number fields; its Hodge-number calculation is performed embedding by embedding.

G2. Integral character congruences. Proposition 4.2 gives the rotation-module isomorphism

$$H^1(C_m, \mathbb{Z}_\ell) \simeq \mathbb{Z}_\ell[\mu_m]^2,$$

including when $\ell \mid m$. Thus evaluation at characters with the same reduction gives isomorphic semisimplified residual representations, as in Lemma 5.3 and Corollary 5.4. In particular the congruences used below do not require integral character projectors with denominator divisible by $3p$.

G3. Two geometric monodromy elements. For a nontrivial residual character with value λ , Theorem 4.5 gives, in one basis, both matrices

$$T_a = \begin{pmatrix} 1 & \sigma c_\lambda \\ 0 & 1 \end{pmatrix}, \quad T_b = \begin{pmatrix} 1 & 0 \\ \tau c_\lambda & 1 \end{pmatrix}, \quad c_\lambda = 2 - \lambda - \lambda^{-1}, \quad \sigma, \tau \in \{1, -1\}.$$

The two loops act simultaneously on all character pieces of each cover.

G4. Compact-type local specialization. The first degeneration has Jacobian E^n . Proposition 7.5 and Corollary 7.6 provide nonempty local opens at every $v \mid 3p$, over arbitrary finite extensions of the relevant $\mathbb{Q}_\ell$, in which every rank-two piece has the same type as E : potentially crystalline ordinary, potentially crystalline with slopes $1/2$, or potentially totally toric with nonzero monodromy.

G5. The second boundary model. On the constant- E parameter cover, let R_* be a boundary point with $Q = [n]R_* \in E[2] \setminus \{O\}$ and $O, \pm R_*, Q$ distinct. The local blowup and normalization calculation of [M], Lemma 3.2, equation (3.7), and Lemma 3.5 gives an unramified degree-three cover of E , possibly disconnected, with Kummer class $-2(n/3)R_*$ and a geometrically rational tail, meeting in three geometric nodes of smoothing order one. The involution extends to this model. Section 12 proves connectedness at exact-order-six R_* , splitness after adjoining $E[6]$, and the quotient's width three; none of those arithmetic conclusions is assumed here. Only the all-CM extension needs this second boundary input.

Local automorphic comparison (LC). The package $\mathcal{G}$ also includes the local-comparison assertion of [M], Lemma 9.2: when a character system has become automorphic, its next local automorphic input has zero monodromy in the potentially good case and is a quadratic twist of Steinberg in the toric case, with the ordinarity matching G4. This assertion uses compatibility at a different coefficient prime, purity, and genericity of cuspidal GL_2 local components. It is a separate source input from G1–G5. No degree-two condition is used in the argument stated there. The simultaneous arithmetic Hilbert specialization over arbitrary CM fields is proved in Section 4, rather than assumed as part of $\mathcal{G}$.

The elliptic identifications from Proposition 5.2 are

$$V_1 = H_{\text{ét}}^1(B_t), \quad V_\eta = H_{\text{ét}}^1(A_t), \quad A_t = \mathrm{Jac}(C_3(t)/\langle w \rangle), \quad \mathrm{ord}(\eta) = 3.$$

The quotient and its Jacobian are defined over k , because $C_3(t)$ and w are. The fact that the source introduces A_t over F does not prevent this earlier descent. A rational point on the genus-one quotient is unnecessary for its Jacobian to be an elliptic curve.

We also use Serre's open-image theorem [Serre], Hilbert irreducibility with weak approximation, Chebotarev, and automorphic Galois representations. The exact base-change and descent result is [ACC], Proposition 6.5.13, which permits both imaginary CM and totally real fields and does not require a polarization hypothesis. Its proof uses cyclic base change and descent from [AC], Chapter 3, Theorems 4.2 and 5.1. The specific automorphy inputs are Caraiani–Newton [CN], Theorem 5.2, Theorem 6.1, and Lemma 6.1.3. Theorem 5.2 allows arbitrary CM fields and ramification at its coefficient prime. Its extra condition in characteristic five will not occur in the three lifts.

3 Normal closures and large residual images

Lemma 3.1. *The normal closure of a CM field is CM, with central complex conjugation. Every subfield of a CM field is either totally real or CM. Consequently, for a CM field K , the extension $K(\mu_N)/K$ is abelian and all its intermediate fields are CM.*

Proof. If c_K is the CM involution, then every embedding $\sigma : K \rightarrow \mathbb{C}$ satisfies $\overline{\sigma(x)} = \sigma(c_K x)$. The normal closure is generated by the conjugates of K , so complex conjugation commutes with every automorphism there. Its fixed field is totally real: every conjugate of an element fixed by this central involution is real. The involution is nontrivial, so the normal closure is CM. The same argument applies to composita of CM fields.

Inside a Galois CM normal closure, central complex conjugation preserves every subfield. Its restriction either acts trivially, giving a totally real field, or makes the subfield a quadratic totally imaginary extension of its totally real fixed field. Apply this to the subfields of the compositum $K(\mu_N)$. A field containing K cannot be totally real. Finally, $K(\mu_N)/K$ is abelian because it is obtained from the abelian extension $\mathbb{Q}(\mu_N)/\mathbb{Q}$ by base extension. $\square$

Let now K be any CM field, $d = [K : \mathbb{Q}]$, and let $\widetilde{K}$ be its *fixed* normal closure. For a non-CM elliptic curve E/K , Serre's theorem allows a prime

$$p > \max(100, 2d + 1), \quad \rho_{E,p}(G_{\widetilde{K}}) \supseteq \mathrm{SL}_2(\mathbb{F}_p).$$

Set

$$F = K(\mu_{3p}), \quad D_0 = \widetilde{K}(\mu_{3p}).$$

The automorphy field will be F , not the normal closure. The latter is used only to control Frobenius conditions at every conjugate.

Lemma 3.2. *For every embedding $\sigma : K \hookrightarrow \widetilde{K}$, $\rho_{E^\sigma,p}(G_{D_0}) = \mathrm{SL}_2(\mathbb{F}_p)$.*

Proof. Normality of $\widetilde{K}/\mathbb{Q}$ transports the large-image assertion to every conjugate E^σ . Since $D_0/\widetilde{K}$ is abelian, the image of G_{D_0} contains the commutator subgroup of the image of $G_{\widetilde{K}}$. The group $\mathrm{SL}_2(\mathbb{F}_p)$ is perfect for $p > 3$; hence that commutator contains $\mathrm{SL}_2(\mathbb{F}_p)$. Conversely, the determinant is trivial over D_0 , which contains μ_p . This proves equality. $\square$

This argument deals with the dependence of D_0 on p . It does not compare p with the degree of a field chosen after p , and does not require a large image after passing to the finite extension cut out by the later Hilbert specialization.

Lemma 3.3. *For every quadratic twist B/K of E , its residual representation at p is absolutely irreducible and decomposed generic over F .*

Proof. On G_{D_0} , form the joint image

$$J \subseteq \prod_{\sigma: K \hookrightarrow \widetilde{K}} \mathrm{SL}_2(\mathbb{F}_p).$$

Each of its d projections is surjective. In $\mathrm{SL}_2(\mathbb{F}_p)$, precisely p^2 elements have trace 2: writing such a matrix as $\begin{pmatrix} a & b \\ c & 2-a \end{pmatrix}$, its determinant condition is $bc = -(a-1)^2$. For $a = 1$ there are $2p-1$ solutions; the other $p-1$ values give $p-1$ solutions each. The count for trace -2 is the same.

The fibers of each surjective projection from J have equal cardinality. The union bound therefore gives

$$\frac{\#\{g \in J : \text{some factor has trace } 2 \text{ or } -2\}}{|J|} \leq \frac{2dp}{p^2 - 1} < 1. \quad (1)$$

No independence of the conjugate residual representations is assumed. Choose an element regular semisimple in every factor. In a finite Galois extension of $\mathbb{Q}$ recording D_0 and the torsion fields, its conjugacy class still has this property, because conjugation permutes the d factors.

Chebotarev gives rational primes q completely split in D_0 , hence in F , with these Frobenius actions. In particular $q \equiv 1 \pmod{p}$. At every place over q , distinct eigenvalues give the

decomposed-generic condition: neither eigenvalue ratio equals q modulo p . Absolute irreducibility over F follows from the full special-linear image on $G_{D_0} \subseteq G_F$. Quadratic twisting preserves irreducibility and eigenvalue ratios; exclude the finitely many primes ramified in the twist. $\square$

4 Simultaneous arithmetic specialization

Lemma 4.1. *Assume $\mathcal{G}$. Prescribe nonempty opens $\Omega_v \subset U(K_v)$ at finitely many finite places, including every $v \mid 3p$. There is $t \in U(K)$ in all these opens for which, over F :*

1. *every nontrivial character space for $(m, \ell) = (3, p), (3, 5), (p, 3)$ is absolutely irreducible on $G_{F(\zeta_\ell)}$ and decomposed generic;*
2. *the same assertions hold for $V_1 \bmod p$;*
3. *A_t is non-CM, and the image of $A_t[5]$ on G_D is $\mathrm{SL}_2(\mathbb{F}_5)$, for the auxiliary Galois field D used below. In particular, if $\mu_5 \subset K$, its full mod-five image is $\mathrm{SL}_2(\mathbb{F}_5)$.*

Proof. Put $W = \mathrm{Res}_{K/\mathbb{Q}} U$, an open of $\mathbb{A}_{\mathbb{Q}}^d$, and choose a finite Galois field $D/\mathbb{Q}$ containing D_0 and ζ_5 ; $D = D_0(\zeta_5)$ suffices. Then

$$W_D \simeq \prod_{\sigma: K \hookrightarrow D} U_D^\sigma.$$

For each listed pair, pull back the finite local system $H^1(C_m, \mathbb{F}_\ell)$ from every factor. Intersect the kernels of these finite actions in $\pi_1(W_D)$, and take the normal core in $\pi_1(W)$. We obtain a finite quotient

$$\pi_1(W) \twoheadrightarrow H \twoheadrightarrow \mathrm{Gal}(D/\mathbb{Q}).$$

In one geometric factor, the two opposite nonidentity unipotents in G_3 have different fixed lines, hence force absolute irreducibility. There is also a single geometric element regular semisimple on *all* its nontrivial characters for each pair. For $m = 3$, $c_\lambda = 3$, and $T_a T_b^2$ has trace 20 or -16 , neither equal to ± 2 modulo 5 or modulo $p > 100$. For $m = p, \ell = 3$, use $T_a T_b$, with trace $2 + \sigma\tau c_\lambda^2$. A failure of regular semisimplicity would imply $c_\lambda^2 = \pm 4$, giving a polynomial of degree at most four for λ over $\mathbb{F}_3$. Its order p would divide $3^e - 1$ for some $e \leq 4$, impossible for $p > 100$.

Geometric loops in different factors act independently on the systems pulled back from those factors. Multiplying the chosen factor loops gives, for each pair, an element $h_{m,\ell}$ of the constant-field kernel which is regular semisimple on every factor and character. The opposite unipotents are also in this kernel.

For each rational prime q ,

$$W(\mathbb{Q}_q) = \prod_{v|q} U(K_v).$$

Thus the local conditions are nonempty opens on $W(\mathbb{Q}_q)$. Hilbert irreducibility with weak approximation produces $x \in W(\mathbb{Q})$ in those opens with surjective specialization $s_x : G_{\mathbb{Q}} \rightarrow H$. This is precisely the arbitrary-dimensional Hilbert statement already recorded as [M], Lemma 8.3. The constant-field map implies

$$s_x(G_D) = \ker(H \rightarrow \mathrm{Gal}(D/\mathbb{Q})). \quad (2)$$

The point x corresponds to one $t \in U(K)$; its coordinates over D are the actual conjugates $\sigma(t)$, not independently selected parameters.

Equation (2) retains the opposite unipotents over G_D , hence also over $G_{F(\zeta_\ell)}$. Since $\ell \nmid m$ for the three pairs, character decomposition of the smaller cover commutes with reduction. The comparison with the C_{3p} -pieces preserves semisimplified reductions.

Conjugation permutes the specialized conjugate covers and their rotation characters. Thus every conjugate of $h_{m,\ell}$ has the required regular-semisimple action. Chebotarev in the extension cut out by s_x gives rational primes split completely in D and regular semisimple at every place above them. They are 1 mod ℓ , establishing decomposed genericity. Each pair may use a different prime. Assertion 2 is Lemma 3.3.

For $m = 3$ and $\ell = 5$, $c_\lambda = 3$. Powers of the two nonzero unipotents generate both root subgroups, hence a group of order 120 isomorphic to $\mathrm{SL}_2(\mathbb{F}_5)$. The rational isomorphism between the character system and $V_5(A_t)$ identifies their semisimplified residual representations. Since the character representation is absolutely irreducible, the actual residual representation on $A_t[5]$ is isomorphic to it. In an $\mathbb{F}_5$ -basis of $A_t[5]$ this subgroup has determinant one and order 120, so it is $\mathrm{SL}_2(\mathbb{F}_5)$. The determinant is trivial on G_D ; if $\mu_5 \subset K$ it is trivial on G_K as well. This proves the asserted full-image statements without requiring an integral isomorphism of the two original lattices.

Finally, $A_t \bmod p$ has the two opposite noncommuting unipotents of order p . A CM elliptic curve has all endomorphisms defined over an extension of degree at most two; its residual image consequently has an abelian normal subgroup of index at most two. Both odd-order unipotents would lie in that subgroup and commute, a contradiction. $\square$

5 The seed field, three lifts, and descent

Theorem 5.1 (Conditional CM scope extension). *Assume $\mathcal{G}$. Let K be any CM number field with $\zeta_5 \notin K$. Every elliptic curve E/K is modular. For non-CM E , there is a cuspidal weight-zero representation π_E of $\mathrm{PGL}_2(\mathbb{A}_K)$ realizing its cohomology at every coefficient prime, with full finite-place local compatibility in the usual Tate normalization.*

Proof. First let E be non-CM. Choose p, F, t as above, using all the opens from G4. The auxiliary elliptic curve A_t is over K . Its residual representation modulo 5 is absolutely irreducible on $G_{F(\zeta_5)}$, hence on $G_{K(\zeta_5)}$. The rational split prime witnessing decomposed genericity over F also witnesses it over K : all relevant local residue degrees are one.

Caraiani–Newton, Theorem 6.1, applies to A_t/K . *This is the only step requiring $\zeta_5 \notin K$.* Base change its modular representation through the abelian extension F/K , using [ACC], Proposition 6.5.13(1). It remains cuspidal. Indeed, in a cyclic prime-degree step, noncuspidal base change would give a nontrivial self-twist by a character of that cyclic extension. A corresponding intertwiner of Galois representations would be scalar after restriction to the upper field, by absolute irreducibility and Schur’s lemma; a scalar cannot implement a nontrivial twist. Irreducibility at every step follows already from the residual irreducibility over F .

Let η, ψ have orders 3 and p . G2 gives

$$V_\eta \equiv_p V_{\eta\psi} \equiv_3 V_\psi \equiv_p V_1 = H_{\text{et}}^1(B_t). \quad (3)$$

Starting with the base change of A_t , apply Caraiani–Newton, Theorem 5.2, successively as follows:

Automorphic input	Coefficient prime	Target
V_η	p	$V_{\eta\psi}$
$V_{\eta\psi}$	3	V_ψ
V_ψ	p	V_1

Determinants, all labeled weights, and potential semistability are G1. Residual hypotheses are Lemmas 3.3 and 4.1. In the potentially good cases G4 and LC give matching ordinarity and

zero automorphic monodromy; in the toric case they give noncrystalline representations and an ordinary Steinberg input. Compatible Frobenius polynomials transport the automorphic realization to the coefficient prime needed for the next lift, using semisimplicity and Chebotarev. Thus all hypotheses in Theorem 5.2 are met. The lifting primes are $p, 3, p$, so its characteristic-five exception never enters. In particular, ζ_5 is allowed to lie in F .

We have modularity of B_t/F . The extension F/K is abelian, with CM intermediate fields by Lemma 3.1. Apply [ACC], Proposition 6.5.13(2). Its argument descends through a tower of cyclic prime-degree extensions: For completeness, if L'/L is one such step and $r|_{G_{L'}}$ is realized by a cuspidal representation Π , extension of r to G_L gives $\text{Gal}(L'/L)$ -invariance of Π by strong multiplicity one. Cyclic descent gives π/L . Its complex archimedean parameters still have weight zero. After identifying restrictions, Schur's lemma gives

$$r_\ell(\pi) = r \otimes \delta, \quad \delta : \text{Gal}(L'/L) \longrightarrow \overline{\mathbb{Q}}_\ell^\times.$$

Twisting by the inverse Hecke character removes δ . Absolute irreducibility holds at every step because it holds over F . No nonsolvable descent and no prime-to- ℓ extension-degree assumption is used.

Twist B_t/K back to E/K . Caraiani–Newton, Lemma 6.1.3, supplies realization at all coefficient primes and

$$\text{WD}(\text{H}_{\text{ét}}^1(E_{\overline{K}}, \mathbb{Q}_\ell)|_{G_{K_v}})^{F\text{-ss}} \simeq \text{rec}_{K_v}^T(\pi_{E,v}) \quad (v \nmid \ell), \quad (4)$$

after the chosen coefficient isomorphism. Choosing ℓ away from each place yields every finite local factor.

For CM E , the classical Hecke-character construction applies over the compositum with its imaginary quadratic CM field M . It gives an isobaric sum if $M \subseteq K$, and quadratic automorphic induction otherwise. $\square$

6 Totally real descent

Corollary 6.1 (Conditional totally real modularity). *Assume $\mathcal{G}$. Every elliptic curve over every totally real number field k is modular of parallel weight two, with matching finite local parameters.*

Proof. For non-CM E/k , choose an odd rational prime $q \neq 5$ unramified in k and set $K = k(\sqrt{-q})$. This is a quadratic CM extension, ramified at places above q . Since $k(\zeta_5)/k$ is unramified at those places, ζ_5 cannot belong to K : otherwise the nontrivial subextension $k(\zeta_5)/k$ of this quadratic extension would equal K/k . Theorem 5.1 gives a modular representation Π/K .

Since E is defined over k , Π is invariant under $\text{Gal}(K/k)$. Apply [ACC], Proposition 6.5.13(2), to $V_\ell(E)$, choosing ℓ so that its restriction to G_K is irreducible. It produces a cuspidal regular algebraic representation π/k realizing $V_\ell(E)$. The real archimedean type requires a check beyond CM-to-CM descent. At a complex place of K its unitary parameter is

$$\chi \oplus \chi^c, \quad \chi(z) = z/|z|.$$

The characters are distinct. Any $W_{\mathbb{R}}$ -parameter restricting to this must interchange the two lines, and is $\text{Ind}_{W_{\mathbb{C}}}^{W_{\mathbb{R}}} \chi$, the parameter of weight-two discrete series. For example, the nontrivial Weil element can act by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. There is no principal-series extension of these two angular characters. Thus π has parallel classical weight two. The character correction implicit in the descent proposition is at most a sign twist at a real place, which preserves this discrete series. Its central character is trivial: its base change is trivial, so it is either trivial or the quadratic character of K/k , and the latter has nontrivial sign at every real place whereas weight-two discrete series has trivial central character.

One can also obtain all finite local parameters using just the CM comparison already invoked. Fix a finite place v of k . Choose a fresh finite place $w \neq v$, away from 2 and 5, and by weak approximation choose $a \in k^\times$ negative at every real place, a square in $k_v^\times$, and of valuation one at w . The extension $L^{(v)} = k(\sqrt{a})$ is CM, splits at v , and avoids ζ_5 by its ramification at w . The modular representation of $E/L^{(v)}$ and $\mathrm{BC}_{L^{(v)}/k}(\pi)$ agree at good places, hence everywhere by strong multiplicity one; base change stays cuspidal by the same irreducibility argument. At v local base change is the identity. Formula (4) over $L^{(v)}$ therefore gives the local parameter over k_v . Repeat for each v . CM curves are covered by classical quadratic automorphic induction. $\square$

7 The exceptional characteristic-five obstruction

The three lifts in (3) use $p, 3, p$, with $p > 5$. Their lifting hypotheses allow ζ_5 in the automorphy field. The root-of-unity restriction in Theorem 5.1 enters solely when the auxiliary curve A_t is first proved modular by CN, Theorem 6.1. Its mod-five seed proposition, Proposition 6.1.5, has no such restriction; its mod-three seed proposition, Proposition 6.1.6, uses the internal 3–5 switch of AKT, Proposition 9.15.

Suppose $\mu_5 \subset K$ and $\bar{\rho}$ has image $\mathrm{SL}_2(\mathbb{F}_5)$. For $M = \mathrm{ad}^0 \bar{\rho}$ one has the exceptional line

$$H^1(\mathrm{SL}_2(\mathbb{F}_5), M) \simeq \mathbb{F}_5. \quad (5)$$

It is not detected by ordinary Taylor–Wiles primes. The image restrictions in AKT, Theorem 8.1, and CN, Theorem 5.2, exclude this global obstruction. Twisting does not change the projective image; soluble restriction preserves full $\mathrm{SL}_2(\mathbb{F}_5)$ image because that group is perfect. Our alternative is to exclude the line from the relevant dual Selmer group using a local ordinary condition. Sections 8–11 are independent of $\mathcal{G}$.

8 The local Tate condition

Let $L/\mathbb{Q}_5$ contain μ_5 , and let T_q/L be a Tate elliptic curve with parameter q . Assume

$$v_L(q) > 0, \quad 5 \nmid v_L(q). \quad (6)$$

Up to a choice of basis, and using either the Tate-module convention or its cohomological twist, its residual representation has the form

$$\bar{\rho}_q(g) = \begin{pmatrix} 1 & a_q(g) \\ 0 & 1 \end{pmatrix}.$$

The character a_q is the Kummer character of q , after choosing an identification $\mu_5 \simeq \mathbb{F}_5$. Since $5 \nmid v_L(q)$, $L(q^{1/5})/L$ is totally ramified of degree 5. In particular the residual inertia image is the full unipotent subgroup $U \simeq C_5$.

Write

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathfrak{n} = \mathbb{F}_5 e, \quad \mathfrak{b} = \mathbb{F}_5 e \oplus \mathbb{F}_5 h.$$

These are G_L -stable subspaces of $M = \mathfrak{sl}_2(\mathbb{F}_5)$.

Lemma 8.1 (Restriction of the exceptional class). *Every nonzero class in $H^1(\mathrm{SL}_2(\mathbb{F}_5), M)$ has nonzero image in $H^1(U, M/\mathfrak{b})$, and hence in $H^1(L, M/\mathfrak{b})$ when $G_L \twoheadrightarrow U$ is the Tate residual representation.*

Proof. The centre has order 2 and acts trivially on M , so we may equally work with A_5 . Restriction to its Sylow 5-subgroup is injective, because restriction followed by corestriction is multiplication by $[A_5 : U] = 12$. For $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$,

$$u(e) = e, \quad u(h) = h - 2e, \quad u(f) = f + h - e.$$

Thus $(u - 1)M = \mathfrak{b}$ and $1 + u + u^2 + u^3 + u^4 = 0$ on M . The cyclic-group formula gives

$$H^1(U, M) = M/(u - 1)M = M/\mathfrak{b}.$$

Projection to the trivial module $M/\mathfrak{b}$ induces an isomorphism on $H^1(U, -)$. Its inflation to G_L is injective. The resulting character is a nonzero scalar multiple of a_q . $\square$

Let $\mathcal{L}_L \subset H^1(L, M)$ be the tangent space of ordinary, fixed-determinant deformations relative to their weight algebra. Its local Tate-dual annihilator is denoted by $\mathcal{L}_L^\perp$.

Lemma 8.2 (The ordinary condition kills the exceptional line). *For the Tate representation satisfying (6),*

$$\mathcal{L}_L = \text{im}(H^1(L, \mathfrak{n}) \longrightarrow H^1(L, M)).$$

Consequently a nonzero exceptional class inflated from $H^1(\text{SL}_2(\mathbb{F}_5), M)$ does not belong to $\mathcal{L}_L^\perp$.

Proof. In an upper-triangular ordinary deformation with fixed determinant and fixed inertial weights, the diagonal infinitesimal character is unramified. Lifting the extension requires its cup product with a_q to vanish. The pairing of a_q with a nonzero unramified character is a nonzero multiple of $v_L(q)$. Thus that diagonal direction is zero. All upper extension directions $H^1(L, \mathfrak{n})$ remain available. Changing the ordinary line only changes the class by a coboundary. This proves the tangent-space assertion; it is also the fixed-determinant form of AKT's calculation in Proposition 4.3.

For the trace pairing on M , one has $\mathfrak{n}^\perp = \mathfrak{b}$. Functoriality and nondegeneracy of local Tate duality therefore imply

$$\mathcal{L}_L^\perp \subset \ker(H^1(L, M(1)) \longrightarrow H^1(L, (M/\mathfrak{b})(1))).$$

Here the mod-5 cyclotomic twist is trivial. Lemma 8.1 shows that the exceptional class has nonzero image under this map. $\square$

Remark. The condition $5 \nmid v_L(q)$ does two jobs. It ensures a ramified unipotent residual image, and makes the ordinary deformation ring smooth relative to weight. Nonzero unipotent image alone is not being substituted for both facts.

9 The ordinary local ring

Let $d = [L : \mathbb{Q}_5]$, and let $\mathcal{O}$ be a sufficiently large coefficient ring. Put $\Lambda_L = \mathcal{O}[(\mathcal{O}_L^\times)^f]$, with $(\mathcal{O}_L^\times)^f$ the maximal torsion-free pro-5 quotient. Thus $\dim \Lambda_L = 1 + d$. We use the weight branch through the Tate representation. The fixed determinant is ϵ_5^{-1} .

Proposition 9.1 (Fixed-determinant form of AKT, Proposition 4.3). *For (6), the ordinary lifting ring is formally smooth over Λ_L , of relative dimension $d + 3$ and total dimension $2d + 4$. Moreover*

$$\dim \mathcal{L}_L = d + 1, \quad h^0(L, M) = 1, \quad \dim \mathcal{L}_L - h^0(L, M) = d.$$

Proof. AKT, Section 4.2.1, is stated for a general prime p under exactly the Kummer-valuation hypothesis (6). Proposition 4.1 proves representability of the ordinary condition and uniqueness of its ordered ordinary line; Proposition 4.2 identifies it with the matrix relations provided by ordinary local–global compatibility. Proposition 4.3 proves formal smoothness, with two weight characters and without fixing the determinant. The ring there has dimension $5 + 3d$ over a weight ring of dimension $1 + 2d$.

At the odd prime 5, first impose the determinant relation on the two inertial weight characters. This removes d weight variables. The remaining unramified determinant character splits off by its unique square root congruent to 1. Fixing it removes one further variable. This gives dimension $2d + 4$ over Λ_L , and relative dimension $d + 3$.

For clarity, the essential infinitesimal argument can be given directly. Write an ordinary lift as

$$\begin{pmatrix} \phi & * \\ 0 & \epsilon_5^{-1}\phi^{-1} \end{pmatrix},$$

so its extension module has character $\epsilon_5\phi^2$. For a small extension of coefficient algebras with kernel I , the obstruction to lifting the extension cocycle lies in $H^2(L, \mathbb{F}_5) \otimes I$, a one-dimensional obstruction space. Changing the unramified part of ϕ by xur changes that obstruction by

$$\pm 2x(ur \smile a_q).$$

Its invariant is a nonzero multiple of $2xv_L(q)$, so every obstruction can be removed. The ordinary line is recovered from an inertia matrix with unit upper-right entry. The forgetful map from the flagged ordinary functor is therefore an isomorphism at this residual representation.

At fixed weight the diagonal infinitesimal direction is forced to vanish. The upper cocycles contribute $h^1(L, \mathbb{F}_5) = d + 2$ parameters and the line contributes one. The coboundary dimension is $3 - h^0(L, M) = 2$, giving $\dim \mathcal{L}_L = d + 1$. In particular the Kummer torsion directions have not been discarded by taking a flat quotient. $\square$

10 The exceptional-five lifting argument

Proposition 10.1 (Proposed exceptional-5 ordinary lifting variant). *Let K be a CM field containing μ_5 . Let $\rho : G_K \rightarrow \mathrm{GL}_2(\mathbb{Q}_5)$ have determinant ϵ_5^{-1} , be unramified almost everywhere, and be ordinary of weight zero. Suppose:*

1. $\bar{\rho}$ is decomposed generic and has image $\mathrm{SL}_2(\mathbb{F}_5)$;
2. $\bar{\rho}$ is the residual representation of a cuspidal, ordinary, weight-zero representation π of $\mathrm{PGL}_2(\mathbb{A}_K)$;
3. at every place above 5, ρ and $r_5(\pi)$ are the cohomology representations of split Tate elliptic curves, with the common residual Kummer class having valuation nonzero modulo 5. In particular their local representations are extensions $0 \rightarrow \mathbb{Q}_5 \rightarrow V \rightarrow \mathbb{Q}_5(-1) \rightarrow 0$, after extending coefficients.

The proposed lifting assertion $\mathcal{T}_5$ is that ρ is automorphic. We give the patching adaptation below as a research draft of this assertion.

We give the full construction as a candidate proof, using the published abstract patching theorem. The assertion is not an application of the excluded case of AKT, Theorem A.14, or CN, Theorem 5.2.

10.1 Preparation and the two deformation problems

Make the soluble CM base changes used before AKT, Theorem A.7, but require the extensions at every 5-adic place to be unramified. This preserves (6), and allows $[K_v : \mathbb{Q}_5] > 4$. The imaginary quadratic auxiliary fields can be chosen to split at 5 and at a protected decomposed-generic rational prime. The latter prime then remains a witness after every extension. Arrange that the automorphic seed is Iwahori-spherical and that, away from 5, the Galois representations are unramified or unipotently ramified, with trivial residual representation at the level-raising places. These are local soluble extension requirements and can be imposed together with the unramified requirements at 5. Full $\mathrm{SL}_2(\mathbb{F}_5)$ image survives soluble restriction. The printed step which trivializes the residual representation at 5 is omitted; Proposition 9.1 supplies its replacement.

Write

$$D = [K^+ : \mathbb{Q}], \quad d_v = [K_v : \mathbb{Q}_5], \quad \Lambda = \widehat{\bigotimes_{v|5, \mathcal{O}} \Lambda_v}, \quad \dim \Lambda = 1 + 2D.$$

We use one inertial weight character at each v , with ordered diagonal characters ϕ_v and $\epsilon_5^{-1} \phi_v^{-1}$. The maximal torsion-free pro-5 weight branch contains the Tate point: under reciprocity the cyclotomic character is the local norm, up to inversion, and is trivial on the 5-power roots of unity of K_v . Let $\mathfrak{p}_0 = \ker(\Lambda \rightarrow \mathcal{O})$ be the weight-zero augmentation.

Let $S = S_5 \cup R$, with R the prepared prime-to-5 ramification set. Use the smooth fixed-determinant ordinary ring of Proposition 9.1 at every place above 5. At $v \in R$, use the fixed-determinant rings of AKT, Lemma A.2, with tame characters $x_v = 1$ or $x_v = \chi_v$, where χ_v has 5-power order and $\chi_v^2 \neq 1$. Such characters exist after the preparation $q_v \equiv 1 \pmod{5}$ and enlarging the coefficient field. The resulting problems are denoted $\mathcal{S}_1$ and $\mathcal{S}_\chi$; they are canonically identical modulo the coefficient uniformizer ϖ . Additional smooth prime-to-5 places allowed by Lemma A.3 cause no change in the argument.

10.2 The retained local condition and Taylor–Wiles primes

Choose $v_0 \mid 5$ and put

$$T = S \setminus \{v_0\}, \quad t = |T|, \quad d_0 = d_{v_0}.$$

The split imaginary quadratic auxiliary extension gives at least two places above 5, so T is nonempty. In the T -framed relative dual Selmer group, places of T have no dual local condition, whereas v_0 retains $\mathcal{L}_{v_0}^\perp$. The relative tangent is taken over the fixed weight algebra; Proposition 9.1 gives

$$\dim \mathcal{L}_{v_0} = d_0 + 1, \quad h^0(K_{v_0}, M) = 1.$$

Lemma 8.2 excludes the inflated exceptional line from this dual Selmer group. The local condition is the same for $\mathcal{S}_1$ and $\mathcal{S}_\chi$. Taking $T = S$ would drop this condition and invalidate the exclusion.

Here is the replacement for AKT, Lemma A.5. Put $K_N = K(\mu_{5^N})$. The residual field is disjoint from the abelian extension K_N/K because $\mathrm{SL}_2(\mathbb{F}_5)$ is perfect. The combined finite image is $\mathrm{SL}_2(\mathbb{F}_5) \times \mathrm{Gal}(K_N/K)$. Since the adjoint module has no $\mathrm{SL}_2(\mathbb{F}_5)$ -invariants, the H^1 of this combined image is precisely the exceptional line. A dual Selmer class vanishing on the kernel of the combined representation would inflate from that line, and hence must be zero.

Any remaining nonzero class restricts to a nonzero equivariant homomorphism from this kernel to M . The span of its image over the residue coefficient field is all of M , by absolute irreducibility of the adjoint module. This span formulation also covers enlarged residue coefficient fields. Choose a lift of $g = \mathrm{diag}(2, 3)$ with trivial 5^N -cyclotomic character. Varying it by kernel elements makes the class nonzero in $M/(g-1)M$. Chebotarev supplies a regular semisimple Taylor–Wiles prime of norm 1 modulo 5^N detecting the class. Degree-one primes may be chosen as in AKT, Lemma A.5; they have the required imaginary quadratic splitting property.

Iteration and padding produce sets Q_N of one fixed size q , with the relative dual Selmer group zero for both problems. Set $Q_0 = \emptyset$. AKT, Proposition A.4, gives presentations with

$$g = q - 3D - 1 + t + d_0 \quad (7)$$

global variables over the partially framed local ring. Here the omitted ordinary place contributes d_0 , each Taylor–Wiles place contributes one, and the complex places contribute $-3D$. Increase q if necessary. Only the presentation-generator bound is used: dual Selmer vanishing does not assert formal smoothness of the augmented global ring.

10.3 Framing algebras and finite-level presentations

Write

$$R_N = R_{S_1, Q_N}, \quad R'_N = R_{S_\chi, Q_N}, \quad \Delta_N = \prod_{v \in Q_N} k(v)^\times (5).$$

Let $\mathcal{B} = \Lambda[[z_1, \dots, z_{4t-1}]]$ be the formal framing algebra. Absolute irreducibility and $T \neq \emptyset$ identify the framed rings, after compatible choices of representatives and coordinates, with

$$R_{S_1, Q_N}^T \simeq \mathcal{B} \hat{\otimes}_\Lambda R_N, \quad R_{S_\chi, Q_N}^T \simeq \mathcal{B} \hat{\otimes}_\Lambda R'_N.$$

These choices can respect both the Taylor–Wiles augmentation and reduction modulo ϖ , as in AKT, Section 6.2. Fixing the determinant of the representation does not impose determinant-one framing matrices: the framing count remains $4t - 1$.

Let $\Lambda_T = \widehat{\bigotimes}_{v \in T, \mathcal{O}} \Lambda_v$, taking $\Lambda_v = \mathcal{O}$ away from 5. The local coefficient rings are

$$R^{\text{loc}} = \Lambda \hat{\otimes}_{\Lambda_T} \left(\widehat{\bigotimes}_{v \in T, \mathcal{O}} R_{v,1} \right), \quad R'^{\text{loc}} = \Lambda \hat{\otimes}_{\Lambda_T} \left(\widehat{\bigotimes}_{v \in T, \mathcal{O}} R_{v,\chi} \right).$$

Thus the weight factor Λ_{v_0} remains present, although its ordinary lifting-ring factor is omitted. Restriction gives maps from these rings to the corresponding T -framed global rings. Define

$$\begin{aligned} R_\infty &= R^{\text{loc}}[[X_1, \dots, X_g]], & R'_\infty &= R'^{\text{loc}}[[X_1, \dots, X_g]], \\ \Delta_\infty &= \mathbb{Z}_5^q, & S_\infty &= \mathcal{B}[[\Delta_\infty]], & \mathfrak{a}_\infty &= \ker(S_\infty \rightarrow \Lambda). \end{aligned}$$

10.4 The five finite-level patching inputs

We verify the inputs of [ACC], Section 6.4.1. Its formal framing algebra is arbitrary; it does not require framing at every ramified place.

1. **Diamond groups.** The group Δ_N is a quotient of Δ_∞ whose kernel is contained in $5^N \Delta_\infty$, because every Taylor–Wiles norm is 1 modulo 5^N . At $N = 0$ the quotient is trivial.
2. **Perfect complexes and their reductions.** Take the complexes constructed before AKT, Proposition A.13, and specialize the full diamond-operator weight algebra Λ_1 to Λ :

$$C_N = F_{\mathfrak{m}, 1, Q_N} \otimes_{\Lambda_1[\Delta_N]} \Lambda[\Delta_N], \quad C'_N = F_{\mathfrak{m}, \chi, Q_N} \otimes_{\Lambda_1[\Delta_N]} \Lambda[\Delta_N].$$

Here $\mathfrak{m}$ is the non-Eisenstein maximal ideal attached to the common residual representation. AKT, Lemma A.10 and its Taylor–Wiles-level construction give perfectness; the non-neat-level input is supplied by Theorem 5.11 using $\mu_5 \subset K$. The tame coefficient systems 1 and χ agree modulo ϖ , giving the required derived identification of C_N/ϖ and C'_N/ϖ . Their finite Hecke algebras, denoted $\mathbb{T}_N$ and $\mathbb{T}'_N$, have the same image after this reduction. Changing the framing set changes none of these complexes or operators.

3. **Global rings and local structures.** The rings R_N, R'_N have their usual $\Lambda[\Delta_N]$ -structures and agree modulo ϖ . The restriction maps just constructed give the required R^{loc} - and R'^{loc} -structures on $\mathcal{B} \hat{\otimes}_{\Lambda} R_N$ and $\mathcal{B} \hat{\otimes}_{\Lambda} R'_N$. They respect the common reduction.
4. **Uniform presentations.** Equation (7) supplies surjections

$$R_{\infty} \twoheadrightarrow \mathcal{B} \hat{\otimes}_{\Lambda} R_N, \quad R'_{\infty} \twoheadrightarrow \mathcal{B} \hat{\otimes}_{\Lambda} R'_N.$$

Choose generators compatibly with $N = 0$ and modulo ϖ : fix generators at level zero, lift them through the augmentation maps, and extend them by generators in the augmentation kernels. Choose a common generating tuple on the reduced rings and lift it to both sides. Nakayama's lemma and the uniform bound give the required presentations.

5. **Deformation maps to the Hecke actions.** The adaptations of AKT, Propositions A.11 and A.13 give surjections

$$R_N \twoheadrightarrow \mathbb{T}_N/I_N, \quad R'_N \twoheadrightarrow \mathbb{T}'_N/I'_N, \quad I_N^{\delta} = (I'_N)^{\delta} = 0$$

for a single δ independent of N . To verify the changed local condition, first adjoin the ordinary U -operators to the Hecke algebra. AKT, Corollary 5.9 supplies the ordered characters, determinant ϵ_5^{-1} , and the characteristic-polynomial and ordered-product matrix identities. Proposition 4.2 identifies those identities with the actual ramified ordinary lifting functor. Descent of the Galois representation to the algebra generated by traces and weights, together with the descent remark following Proposition 4.2, descends this representable local condition to the smaller Hecke algebra modulo its intersected nilpotent ideal. The determinant already has the required value. This is the argument of AKT, Theorem 6.3 and Corollary 6.4, with its PGL_2 form in Corollary 5.9. The nilpotence bound stays uniform. Frobenius traces and the weight/diamond structures determine the maps, hence give their compatibility modulo ϖ . A genuine action of R_N on the complex before passage through nilpotent ideals is not being asserted.

AKT, Lemma A.12 and the construction after it also give

$$C_N \otimes_{\Lambda[\Delta_N]}^{\mathbf{L}} \Lambda \simeq C_0, \quad R_N \otimes_{\Lambda[\Delta_N]} \Lambda \simeq R_0,$$

with corresponding primed identities. Killing the diamond characters makes the Taylor–Wiles local representations unramified. The identities respect reduction modulo ϖ , the Hecke maps, and the chosen presentations; these are the augmentation requirements of [ACC], Section 6.4.1.

10.5 Patched objects and the support hypotheses

The abstract construction now produces perfect complexes C_{∞}, C'_{∞} over S_{∞} , finite Hecke algebras in their derived endomorphism rings, and compatible surjections from R_{∞}, R'_{∞} modulo uniformly nilpotent ideals. It supplies the S_{∞} -algebra structures on the patched rings. ACC+, Proposition 6.4.17 gives the identified reductions and their compatible ring actions. Lemma 6.4.15 and Proposition 6.4.16 give

$$R_{\infty}/\mathfrak{a}_{\infty} \twoheadrightarrow R_0, \quad C_{\infty} \otimes_{S_{\infty}}^{\mathbf{L}} S_{\infty}/\mathfrak{a}_{\infty} \simeq C_0, \tag{8}$$

and compatibility with the Hecke maps. Only the indicated surjection of rings is needed.

Here are all three hypotheses of ACC+, Assumption 6.3.6.

Dimensions. Since $\sum_{v|5} d_v = 2D$ and each prime-to-5 local factor has relative dimension three, one has

$$\begin{aligned}\dim R^{\text{loc}} &= 1 + 4D - d_0 + 3t, \\ \dim R_\infty &= D + 4t + q, \\ \dim S_\infty &= 2D + 4t + q.\end{aligned}$$

The same formulas hold for the primed ring. The smooth factors are $\mathcal{O}$ -flat. The dimension and component statements of AKT, Lemma A.2, together with the completed tensor-product calculation, give

$$\dim(R_\infty/\varpi) = \dim R_\infty - 1, \quad \dim(R'_\infty/\varpi) = \dim R'_\infty - 1.$$

Hence

$$\dim S_\infty - \dim R_\infty = D. \tag{9}$$

Relative to the printed $T = S$ calculation, deleting R_{v_0} removes $d_0 + 3$ variables over weight, while (7) adds $d_0 - 1$. Both patched dimensions therefore fall by four.

Components. The retained 5-adic local factors are formal power series rings over their weight factors; the omitted place contributes its smooth weight algebra. They introduce no extra components. The prime-to-5 level-raising factors are exactly those in AKT, Lemma A.2. Its first part gives the unique generic component above each maximal generic special-fibre component on the unprimed side; its second part gives the unique characteristic-zero generic component on the primed side. The completed tensor-product and power-series argument at the end of AKT, Theorem A.7 applies with these same factors. The remaining factors are smooth, so the required component and specialization conditions hold. In this all-Tate problem the factors are equidimensional; every characteristic-zero target point lies on a component of maximal dimension.

Generic cohomological concentration. The prime $\mathfrak{p} = (\mathfrak{a}_\infty, \mathfrak{p}_0)$ of S_∞ has one-dimensional characteristic-zero quotient. By (8), specialization there and inversion of 5 gives the weight-zero ordinary localized PGL_2 complex at level zero. The seed π supplies nonzero rational cohomology. AKT, Theorem 5.10 concentrates this cohomology in an interval of length D , up to the degree reversal for the chain/dual convention. This is precisely Assumption 6.3.6(3) with $l_0 = D$; no integral amplitude vanishing is assumed.

10.6 The target point and automorphy

The target representation defines $f : R_0 \rightarrow \mathcal{O}$ at weight zero. Pull $\ker f$ back under the surjection in (8), and denote the resulting prime of R_∞ by $\mathfrak{r}$. Its contraction to S_∞ is $(\mathfrak{a}_\infty, \mathfrak{p}_0)$ and it lies on a maximal component. ACC+, Proposition 6.3.8 and Corollary 6.3.9 therefore place it in the support of

$$H^*\left(C_\infty \otimes_{S_\infty}^{\mathbf{L}} S_\infty/(\mathfrak{a}_\infty, \mathfrak{p}_0)\right)[1/5].$$

Proposition 6.4.16 and the compatible Hecke maps transfer this support to the rational weight-zero ordinary complex at level $U(1, 1)$. By AKT, Theorem 5.10(3), the corresponding eigensystem comes from a cuspidal ordinary weight-zero automorphic representation. Frobenius traces and absolute irreducibility identify its Galois representation with ρ . Finally [ACC], Proposition 6.5.13 descends automorphy through the prepared soluble extension, whose 5-adic ramification was explicitly controlled. This completes the candidate proof of $\mathcal{T}_5$.

11 A modular seed preserving the Tate condition

The following is a local-preservation variant of AKT, Proposition 9.13. It uses AKT's unconditional seed construction, independently of Proposition 10.1.

Lemma 11.1 (Preserving the completions at 5). *Let A/K be an elliptic curve over a CM field, with split multiplicative reduction at every $v \mid 5$ and local Tate parameters q_v . Put $\bar{r} = A[5]$, with its Weil pairing. Then, after a solvable CM extension which splits completely at the 5-adic places, there is a modular elliptic curve B with a symplectic isomorphism $B[5] \simeq \bar{r}$. Its Tate parameters at 5 can be chosen in prescribed sufficiently small neighbourhoods of the q_v . A finite avoidance field and a chosen decomposed-generic prime can also be retained.*

Proof. In AKT's proof of Proposition 9.13, the initial solvable CM extension trivializes the prescribed residual representation at 2, 3 and 5. Triviality at 5 is used to supply local Tate points on the twist $Y_{\bar{r}}$ of $Y(5)$. Here the local curves A_{K_v} with their tautological level structures already provide points on this fixed symplectic twist. Trivialize only at 2 and 3, arrange their splitting over the maximal real subfield and the required unramified i -extension at 2, and keep the places above 5 completely split.

The curve $Y_{\bar{r}}$ is a nonempty open of a projective line. AKT, Lemma 9.7, therefore applies with the prescribed local 5-adic opens, as well as the opens at 2 and 3 used in Proposition 9.13. It gives an elliptic curve B with full and decomposed-generic mod-3 image, the required discriminant conditions at 2 and 3, and the specified mod-5 representation. AKT, Proposition 9.12, proves B modular. That proposition imposes no condition at 5 and no exclusion of ζ_5 .

The internal extension used to prove B modular need not preserve the 5-adic parameters: modularity is descended back to the field on which B was constructed before our proposed 5-adic lifting argument is used. $\square$

The preservation condition is on the local moduli points of $Y_{\bar{r}}$, including their symplectic level structures. In particular the approximation takes place on the prescribed residual twist.

12 Tate specialization of the geometric family

Return to the covers of family 030. After pulling the parameter line back to the fixed elliptic curve E , write the moving point as R and take $n = 3p$. Choose a point R_* of exact order 6. Then

$$[n]R_* = [3p]R_* \in E[2] \setminus \{0\}, \quad -2(n/3)R_* = -2pR_*$$

has exact order 3. This is enough for the degree-3 version of the second degeneration: full $6p$ -torsion is unnecessary.

For the normalized covering function on the constant- E parameter cover, write $f_{j,R}$ for the rational function, over the field obtained by adjoining R , with

$$\operatorname{div}(f_{j,R}) = ([j]R) - (-[j]R) - j(R) + j(-R), \quad f_{j,R}(O) = 1.$$

These are the functions of [M], Lemma 3.1. They satisfy the exact identity

$$f_{ab,R} = f_{a,[b]R} f_{b,R}^a.$$

It follows by comparing divisors and the normalization $f(O) = 1$. For $a = 3$ and $b = p$, the change $Y \mapsto Y/f_{p,R}$ identifies the degree-3 cover for $n = 3p$ with the degree-3 family for $n = 3$ pulled back by $[p]$. The identity is compatible with w . Since $[p]R_* = \pm R_*$, the boundary splitting data can be fixed independently of p .

Lemma 12.1 (Width three at the elliptic endpoint). *After a fixed extension of a characteristic-zero local ground field splitting the degeneration data, the genus-one quotient $C_3/\langle w \rangle$ has an I_3 degeneration at R_* . For a local parameter s there,*

$$q_A(s) = as^3(1 + O(s)), \quad a \neq 0. \quad (10)$$

In particular, at a place of residue characteristic 5, arbitrarily small nonzero specializations can satisfy $5 \nmid v(q_A(s))$.

Proof. The degree-3 special cover consists of a connected unramified elliptic cover $E' \rightarrow E$ and a rational component joined by three nodes, each with smoothing order one. The connectedness follows from the exact-order-3 divisor class $-2pR_*$. The involution w preserves both components, acts by inversion on the elliptic component, and acts on the three node labels by $i \mapsto -i$. There is one fixed node and one pair.

At the fixed node, w acts by $(-1, -1)$ on the two branch parameters. If $xy = s$ is the smoothing equation, the quotient has equation $XY = s^2$. The pair of nodes yields one quotient node of thickness one. The two quotient components are rational. Resolving the thickness-two node inserts one component, yielding the three-component cycle I_3 . Thus the quotient Jacobian has j -invariant with a pole of order 3, which gives (10).

For the splitting field, it is enough to adjoin $E[6]$. The point $(O, 1)$ makes $E' \rightarrow E$ a 3-isogeny whose kernel μ_3 is rational. Its restriction $E'[2] \rightarrow E[2]$ is an isomorphism of Galois modules. If $Q = [n]R_*$, its unique 2-torsion lift Q' is therefore rational, and the three attaching points are $Q' + \ker(E' \rightarrow E)$. They are all rational. The rational tail and the quotient cycle are then split as well.

In the arithmetic local field, the constant a need not be an integer unit. For sufficiently small s , however,

$$v(q_A(s)) = v(a) + 3v(s).$$

Choose arbitrarily large integers $k = v(s)$ avoiding the single residue class $-v(a)/3$ modulo 5. Splitness is preserved in a sufficiently small neighbourhood, for example by the square condition on $-c_6$ in a local Weierstrass model. Finally $t = x(R)$ is étale at the exact-order-6 point R_* , so these conditions define nonempty opens in the original parameter line over the extended local field. $\square$

The extension splitting the local boundary data must be chosen before the parameter is specialized. Taking an arbitrary extension afterwards could multiply the valuation by 5 and invalidate (6). Such local extensions can be accommodated in a solvable CM extension. Here one can make the approximation explicit: take, at each 5-adic place of K^+ , the Galois compositum containing the relevant completions of K , their 6-torsion and their conjugates. The preceding proof shows this already splits the boundary. Its Galois group has order divisible only by 2 and 3, so it is solvable. Realize successive cyclic steps by the degree-2 and degree-3 cases of Grunwald–Wang, keeping every real place and any prescribed auxiliary places split. Taking a Galois closure preserves total reality, solvability, and the prescribed complete splitting. Composing that totally real extension with K gives the required CM extension. The local character theorem used here is recorded, for example, in Conrad, Proposition A.3 [Conrad]; there is no Wang exception for the cyclic steps of degree 2 or 3.

There is no circularity in the choice of p : the 6-torsion preparation is independent of p , so one can fix the solvable CM field first and then choose p large enough for the original residual-image estimates. The variable character pieces use the independent-factor Hilbert argument over the fixed field, not a new degree-dependent union bound for p .

13 Completion of the CM argument

The geometric package is $\mathcal{G}$ from Section 2; the additional split Tate specialization is Lemma 12.1. We now combine it with $\mathcal{T}_5$ from Proposition 10.1.

Corollary 13.1 (Consequence of the proposed lifting variant). *If Proposition 10.1 is validated, then $\mathcal{G}$ implies the modularity of every elliptic curve over every CM number field, including fields containing ζ_5 .*

Proof. The case $\zeta_5 \notin K$ is Theorem 5.1. For the remaining case, elliptic curves with complex multiplication are treated by Hecke characters. For a non-CM curve, first take a solvable CM extension admitting the 6-torsion local boundary data and then choose a large $p > 5$ with the residual properties needed for (3). Impose:

- the compact-type local opens at places above $3p$, as in the original switch, so that its successive representations have matching local types;
- at every place above 5, the second-degeneration opens of Lemma 12.1, so that A_t is Tate and $5 \nmid v(q_{A_t})$;
- the finite residual-image and decomposed-generic conditions on $A_t[p]$, $A_t[5]$, and the required order- p character pieces modulo 3.

Weak approximation together with Hilbert irreducibility gives a common specialization. The 5-adic conditions do not conflict with the opens at 3 and p , or with full global mod-5 monodromy. Apply Lemma 11.1 to $A_t[5]$, preserving the 5-adic completions, the residual avoidance fields and decomposed-generic primes. It supplies a modular B with the same residual representation and Tate type at 5. Proposition 10.1 would then prove A_t modular. Solvable descent returns this modularity to the field used for specialization.

Now pass to $F = K'(\mu_{3p})$, where K' denotes that field. The three ordinary/potentially Barsotti–Tate lifting steps in (3) use CN, Theorem 5.2, only at $p, 3, p$. Its exceptional-5 condition never enters these steps. They give the modularity of B_t , hence of E over F . Abelian cyclotomic descent and then solvable descent give modularity over the original K . $\square$

14 An independent residual-mod-three consequence

The proposed lifting variant has a consequence that does not use $\mathcal{G}$.

Corollary 14.1 (Conditional extension of CN, Theorem 6.1(1)). *Assume Proposition 10.1. Let F be any CM number field and let E/F be an elliptic curve. If $E[3]$ is decomposed generic and $E[3]|_{G_{F(\zeta_3)}}$ is absolutely irreducible, then E is modular. In particular this conclusion would no longer require $\zeta_5 \notin F$.*

Proof. Only $\mu_5 \subset F$ is new. In the proof of AKT, Proposition 9.15, and CN, Proposition 6.1.6, first make the soluble CM preparation that trivializes the prescribed mod-3 representation at the places above 5. The local fields then contain μ_3 , by the determinant condition. On the prescribed twist of $Y(3)$, choose at such a place a Tate curve with parameter

$$q = a^3, \quad v(a) > 0, \quad 5 \nmid v(a).$$

Its mod-3 representation is trivial, and $5 \nmid v(q)$. These conditions hold on a nonempty local open of the twist. The prescribed ordinary, supersingular, or Tate opens at 3 in CN, Proposition 6.1.6, can be retained independently. AKT, Lemma 9.7, constructs a curve C satisfying all these opens, with full and decomposed-generic mod-5 image, including the required finite avoidance condition.

Apply Lemma 11.1 to $C[5]$ and then Proposition 10.1 over the resulting soluble CM extension, preserving its 5-adic completions, full mod-5 image and generic prime. Descend the modularity of C to the field on which it was first constructed. This removes the root-of-unity restriction from the mod-3 seed construction without changing its avoidance or prescribed 3-adic local-type conclusions. Finally CN, Theorem 5.2, at $p = 3$ proves the modularity of the original E , exactly as in Theorem 6.1(1). The irreducibility and decomposed-generic hypotheses of that theorem are retained; this is not yet all-CM modularity by itself. $\square$

15 Scope of the new argument

The field extension in Sections 2–6 uses the source package $\mathcal{G}$ and published automorphy theorems. The extra arithmetic use of the second geometric degeneration is its width three, which is invertible modulo five:

$$\begin{aligned} \text{width-three degeneration at five} &\implies \text{Tate parameter of valuation prime to five} \\ &\implies \text{smooth ordinary local condition} \\ &\implies \text{exclusion of the exceptional adjoint class.} \end{aligned}$$

The condition is imposed on the auxiliary elliptic curve. The original curve need not be ordinary or multiplicative at five. The local ring is an existing input; retaining its dual local condition in the partially framed patching problem is the new global argument.

Independent review should concentrate on Proposition 10.1, especially the partially framed patched complexes, their Galois action through the Hecke algebra, and the support comparison. The local cohomology, local-ring calculation, preservation of the Tate class through the modular seed, and width-three specialization have explicit proofs above. A separate verification of $\mathcal{G}$ remains necessary to certify the source manuscript itself. These dependencies are retained in each main statement.

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